
Domain Shift in Limit Order Book Forecasting

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Introduction

Problem. We study *mid-price movement prediction* from high-frequency limit order book (LOB) time series. Given a sequence of recent LOB states, the task is to predict whether the future mid-price will move up, down, or remain stationary over a fixed horizon. This problem is a canonical benchmark in financial time-series modeling and has been widely used to evaluate deep architectures that operate on multivariate, event-driven sequences with strong temporal dependencies and microstructure noise.

Despite its apparent simplicity, models must simultaneously capture short-term temporal patterns, cross-level interactions, and market microstructure effects, while remaining robust to changes in market conditions. As a result, strong performance typically requires large volumes of labeled high-frequency data, which are expensive, proprietary, and limited in temporal coverage.

Why domain shift is central. A key obstacle in applying deep learning to LOB data is *domain shift*. Real-world LOB datasets such as FI-2010 are small by modern deep learning standards and exhibit variability across assets, days, and market regimes. Models trained on one distribution often degrade sharply when evaluated on another.

In this paper we study *cross-market* shift: the change in distribution of LOB states over time between different exchanges (e.g. Finnish and US stock markets). This is a superset of cross-asset shift (since assets are tied to a particular market) and thus poses a particularly challenging setting for the forecasting task across regimes with varying macro and micro structures. A related and increasingly relevant axis of shift arises from the use of synthetic LOB data: while generative models offer an appealing way to mitigate data scarcity, the gap between synthetic and real distributions is itself a form of domain shift that must be characterized and controlled before such data can serve as a substitute for real target-market labels.

Motivation and hypothesis. Prior work, including DeepLOB, demonstrates that high-capacity architectures can achieve strong in-domain performance but often require extensive training data to do so. For target markets where labeled data is scarce, two complementary substitutes have recently become available: high-fidelity synthetic LOB generators and large pretrained time-series foundation models, both of which promise transferable representations without requiring in-distribution supervision. We hypothesize that one or both can close the cross-market gap and enable practical deployment in markets with limited data. Our empirical study finds that cross-market degradation is severe but largely recoverable from very small amounts of real target data; current synthetic and foundation-model substitutes, however, address it only marginally.

1 Background, Related Work, and Baseline Models

1.1 Limit Order Books

A *limit order book* (LOB) is the central mechanism by which most modern electronic exchanges organize trading in a single asset. Participants interact with the book by submitting orders of two types: a *limit order* specifies a

maximum (buy) or minimum (sell) price at which the trader is willing to transact and a quantity of shares; a *market order* requests immediate execution at the best available price. Unexecuted limit orders accumulate in the book, partitioned into a *bid side* (buy orders) and an *ask side* (sell orders). Orders on each side are ranked by *price-time priority*: the most aggressively priced orders are matched first, with ties broken by arrival time.

At any instant, the highest-priced bid is the *best bid* p_1^b and the lowest-priced ask is the *best ask* p_1^a . The *mid-price* is defined as

$$p_{\text{mid}} = \frac{p_1^a + p_1^b}{2}, \quad (1)$$

and the *spread* is $s = p_1^a - p_1^b > 0$. A transaction occurs whenever an incoming order’s price crosses the opposite side of the book: a buy market order (or a sufficiently aggressive buy limit order) executes against the best ask, and vice versa. Figure 1 illustrates a snapshot of a limit order book at a single point in time.

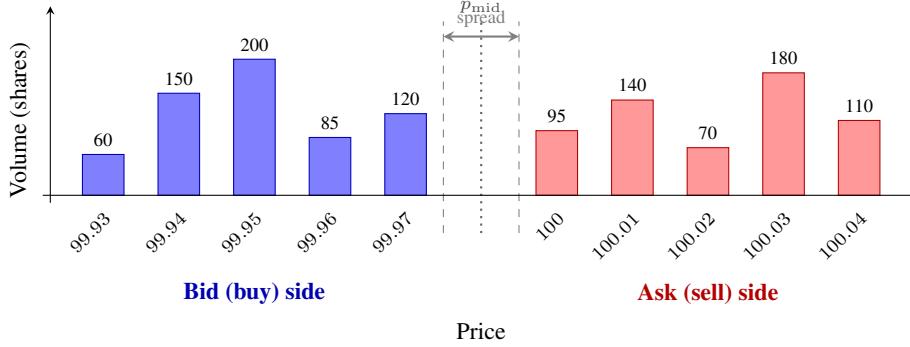


Figure 1: Snapshot of a limit order book at a single point in time, showing five price levels on each side. Each bar represents the total volume of resting limit orders at that price level. The best bid ($p_1^b = 99.97$) and best ask ($p_1^a = 100.00$) define the spread; the mid-price lies at their average.

Message stream representation. The LOB evolves through a discrete sequence of *events* (or *messages*). Each event $e_k = (\tau_k, \omega_k, \pi_k, \nu_k, d_k)$ records a timestamp τ_k , an event type $\omega_k \in \{\text{SUBMIT, CANCEL, EXECUTE}\}$, a price π_k , a volume ν_k , and a direction $d_k \in \{\text{BUY, SELL}\}$. A *submission* adds a new limit order to the book at price π_k ; a *cancellation* removes a resting order; an *execution* (or *trade*) matches an incoming order against the best quote on the opposite side. The ordered stream $(e_1, e_2, \dots)$ fully determines the LOB state at every point in time and constitutes the finest-grained representation of market activity. Both the LOBSTER dataset and the LOBS5 synthetic generator produce data in this message-level format.

Price-volume snapshot representation. While the message stream captures the full generative process, most predictive models—including DeepLOB [18] and the architectures studied in this work—operate on a coarser *snapshot* representation that aggregates resting volume at each price level.

At each observation time t , a depth- D LOB snapshot records the prices and volumes for the top D levels of each side. Concretely, let $p_i^a(t)$ and $v_i^a(t)$ denote the price and aggregate volume at the i -th best ask level, and $p_i^b(t)$ and $v_i^b(t)$ the corresponding quantities on the bid side, with the convention

$$p_1^a(t) < p_2^a(t) < \dots < p_D^a(t), \quad p_1^b(t) > p_2^b(t) > \dots > p_D^b(t). \quad (2)$$

The LOB state at time t is then represented as the vector

$$\mathbf{x}(t) = (p_1^a(t), v_1^a(t), p_1^b(t), v_1^b(t), \dots, p_D^a(t), v_D^a(t), p_D^b(t), v_D^b(t)) \in \mathbb{R}^{4D}. \quad (3)$$

In all experiments in this work we use $D = 10$, yielding a 40-dimensional feature vector at each timestamp. This is the standard input format used by the FI-2010 benchmark [11] and adopted by DeepLOB [18].

To construct the input to a temporal model, we concatenate T consecutive snapshots into a matrix $\mathbf{X}(t) = [\mathbf{x}(t - T + 1); \dots; \mathbf{x}(t)] \in \mathbb{R}^{T \times 4D}$, which serves as the input window for predicting the mid-price movement at a future horizon.

Mid-price movement prediction. The forecasting task studied in this work is a three-class classification problem. Given the current LOB state, we predict the direction of the mid-price change over a future horizon of k ticks. Following the labeling convention of FI-2010 [11], we first compute a smoothed future mid-price as the mean over the next k observations,

$$\bar{p}(t; k) = \frac{1}{k} \sum_{i=1}^k p_{\text{mid}}(t + i), \quad (4)$$

and then assign a label by comparing the relative change to a threshold $\alpha > 0$:

$$y(t; k) = \begin{cases} \text{UP} & \text{if } (\bar{p}(t; k) - p_{\text{mid}}(t))/p_{\text{mid}}(t) > \alpha, \\ \text{DOWN} & \text{if } (\bar{p}(t; k) - p_{\text{mid}}(t))/p_{\text{mid}}(t) < -\alpha, \\ \text{STATIONARY} & \text{otherwise.} \end{cases} \quad (5)$$

The threshold α controls the sensitivity of the labeling: smaller values produce more directional labels at the cost of increased noise, while larger values yield a dominant stationary class. In FI-2010, labels are pre-computed for horizons $k \in \{10, 20, 30, 50, 100\}$ ticks.

1.2 Forecasting Models and DeepLOB

Early approaches to LOB forecasting relied on classical statistical models such as ARIMA and vector autoregressive models. Subsequent work moved toward data-driven methods, including SVM classifiers with engineered LOB features [8], Bag-of-Features and Neural-BoF representations [12], tensor-based models such as the Temporal Attention-augmented Bilinear (TABL) network [16], and deep architectures including CNNs applied directly to LOB snapshots [17] and LSTMs trained on large multi-stock corpora [13]. Among these, the CNN-LSTM hybrid introduced by DeepLOB [18] remains a widely adopted reference point and is the architecture we use as our LOB-specific benchmark throughout this work.

DeepLOB. DeepLOB [18] is a hybrid convolutional–recurrent network designed to predict short-horizon mid-price movements directly from raw LOB snapshots, without hand-crafted features. The input to the model is the matrix $\mathbf{X}(t) \in \mathbb{R}^{T \times 4D}$ defined in Section 1.1, with $T = 100$ recent snapshots and $D = 10$ price levels per side, yielding a 100×40 tensor. The architecture comprises three stages.

(i) *Convolutional feature extraction.* A stack of convolutional layers progressively summarizes the spatial structure of the order book. The initial layer applies (1×2) filters with stride (1×2) , combining the price and volume at each level into a single feature; a subsequent (1×2) stride- (1×2) layer merges adjacent ask and bid columns, producing representations analogous to the micro-price and order-flow imbalance features previously identified as informative microstructure signals. A final (1×10) filter aggregates information across all ten levels, leaving a sequence of feature maps with the temporal dimension preserved through zero-padding. Leaky ReLU activations are used throughout, and no pooling is applied within the convolutional stack so as to avoid smoothing out the temporal location of microstructure events.

(ii) *Inception module.* The resulting feature maps are passed through an Inception module [15] that applies parallel convolutions of different temporal extents (e.g., 3×1 and 5×1) together with a max-pooling branch, then concatenates the outputs. The motivation, by analogy with multi-horizon moving averages in technical analysis, is to allow the network to learn appropriate temporal decay rates rather than fixing them *a priori*.

(iii) *Recurrent classifier.* The Inception output sequence is fed into a 64-unit LSTM layer, and the final hidden state is projected through a softmax to produce class probabilities over $\{\text{UP}, \text{STATIONARY}, \text{DOWN}\}$. Substituting an LSTM for a fully-connected classifier both captures longer-range temporal dependencies in the extracted features and substantially reduces the parameter count, with the full network containing only on the order of 60k parameters.

On FI-2010, DeepLOB outperforms prior baselines—including BoF, N-BoF, CNN-I, and TABL—across all reported horizons, achieving F1 scores in the high 70s to low 80s depending on horizon and evaluation setup [18]. More importantly for our purposes, the original work reports that a model trained on a subset of London Stock Exchange equities maintains comparable accuracy when applied to held-out tickers it has never seen during training. The authors interpret this as evidence that DeepLOB extracts *universal* features of price formation, an attractive property for transfer across instruments.

This claim is directly relevant to the present work. Cross-asset transfer within a single market is a comparatively benign form of distribution shift, since the underlying market microstructure is held fixed. Under cross-market shift, however, we observe in Section 4 that the ‘vanilla’ DeepLOB model performs poorly on forecasting when deployed in a new market.

1.3 Synthetic Limit Order Book Data

High-quality LOB data is costly, proprietary, and limited to a handful of liquid markets, motivating substantial interest in synthetic alternatives for model training and augmentation, backtesting, scenario analysis, and as world models for reinforcement-learning trading agents [2]. The literature has converged on three broad methodologies. *Agent-based simulations* populate a virtual exchange with heterogeneous trading agents whose interactions emergently produce order flow [2]. *Parametric stochastic models*, exemplified by the Cont–Stoikov–Talreja framework [5], specify arrival, cancellation, and execution processes (typically Poisson, with empirically fitted rates) directly at the level of order book events. More recently, *generative AI* approaches have applied

deep generative models—ranging from (conditional) GANs [4] to autoregressive sequence models—to learn order-flow distributions directly from historical message data.

Within this last category, LOBS5 [10] is an end-to-end, token-level autoregressive generative model of LOB message flow. Each LOBSTER-format message (price, size, direction, type) is converted into a sequence of tokens by grouping successive digits, in the spirit of subword tokenization used by large language models. These tokens are modeled by a stack of simplified structured state-space (S5) layers [14], which efficiently capture long-range dependencies in the message stream while jointly conditioning on the evolving order book state. Generated messages are passed through the JAX-LOB simulator [6], which deterministically updates the book and feeds the new state back into the model, yielding a closed-loop, message-by-message simulator of the exchange. Trained on NASDAQ LOBSTER data, LOBS5 achieves low out-of-sample perplexity and produces mid-price returns whose conditional dynamics correlate strongly with the data, providing synthetic order flow at the granularity of individual messages.

We adopt LOBS5 as our source of synthetic data because it has been shown to be the strongest publicly available LOB generator. The recent LOB-Bench framework [9], which evaluates distributional realism through batteries of conditional and unconditional statistics together with market-impact response functions, finds that LOBS5 attains the lowest overall scores—outperforming the parametric Cont et al. baseline, the Coletta et al. conditional GAN [4], and RWKV-based autoregressive transformers—and is uniquely able to reproduce empirical price-impact curves. We use the scaled-up version of LOBS5 introduced alongside LOB-Bench, which expands the original 6.3M-parameter model of Nagy et al. to roughly 35M parameters trained on the full year of 2022 [9]. Because running the generator at scale was prohibitively computationally expensive, we do not perform inference ourselves; instead, we use the synthetic message data released by the authors, which is faithful to real microstructural dynamics and thus well suited for studying cross-market distribution shift.

2 Datasets and Cross-Market Shift

FI-2010 (Finnish Benchmark). The FI-2010 dataset [11] is the most widely used public LOB benchmark, and it serves as the ‘training’/in-distribution domain for our experiments. It contains 10 consecutive trading days (June 1–14, 2010) of order book data from five Finnish stocks. At each timestamp, the dataset records prices and volumes for the top 10 levels of both the bid and ask sides. The data is event-driven and downsampled at intervals of 10 order events. Labels for mid-price movement direction (up, down, stationary) are pre-computed at five prediction horizons: 10, 20, 30, 50, and 100 ticks. We follow the standard split of 7 days for training and 3 days for testing, although in the cross-market setting we train on the full 10 days of FI-2010 before transferring to LOBSTER.

LOBSTER (NASDAQ US Equities). The LOBSTER dataset [7], hosted on HuggingFace¹, provides reconstructed LOB data for major US equities including AAPL, AMZN, GOOG, INTC, MSFT. The only publicly available LOBSTER data is only for the day of June 21st, 2012 — still > 1 million trading events — which we use as our ‘test’/out-of-distribution domain. For each stock, LOBSTER supplies two file types: (i) a *message file* recording every limit order submission, cancellation, and execution with timestamps, and (ii) an *orderbook file* capturing the full book state (prices and volumes at each level) immediately after each event. Data is available at multiple depth levels (1, 5, 10, 30, and 50 levels).

LOBS5 (Synthetic Data). As described in Section 1.3, we use samples of synthetic limit-order-books generated the scaled-up version of LOBS5 [9] as our synthetic LOB data throughout this notebook. The authors provided 3.67 GB of pre-generated samples of LOBSTER-style message streams, from which we use a sample to reconstruct limit-order-book synthetic snapshots. This dataset, as discussed in [9], matches the shape of LOB data well; however, an extension of our methods would be to use the LOBS5 model to generate samples directly and thus attempt to match our target distribution on finetuning transfer (see Section 5).

2.1 Domain Shift

Cross-market deployment of an LOB forecasting model is only practically attractive if the source and target distributions overlap meaningfully. Before evaluating any transfer methodology, we therefore quantify the distributional gap between FI-2010, LOBSTER, and the LOBS5 synthetic data. Our goal in this subsection is twofold: (i) to characterize the shape and magnitude of the shift between in-distribution (FI-2010) and out-of-distribution (LOBSTER) data, and (ii) to assess whether LOBS5 produces samples whose distribution lies closer to either real dataset, motivating its use as an intermediate domain in our adaptation experiments.

Features and methodology. For each dataset we compute five features chosen to span the principal axes along which microstructure can differ: log mid-price returns (`mid_return`), the bid-ask spread (`spread`), top-of-book

¹<https://huggingface.co/datasets/totalorganfailure/lobster-data>

order imbalance (`order_imbalance`, defined as $(v_1^b - v_1^a)/(v_1^b + v_1^a)$), and total resting volume aggregated over the top ten levels on each side (`total_bid_depth`, `total_ask_depth`). Because FI-2010 is distributed in z-score-normalized form while LOBSTER and LOBS5 contain raw quantities, naive comparison of marginal distributions would be dominated by scale and location differences arising from the normalization rather than from the underlying market dynamics. To isolate shape divergence, each feature is standardized *per dataset* using its own median and interquartile range before any cross-dataset comparison; each standardized feature is then subsampled to 80,000 points (seed 42) to ensure a uniform sample size. We then compute (a) the Kolmogorov–Smirnov (KS) two-sample test statistic, (b) the Kullback–Leibler (KL) and Jensen–Shannon (JS) divergences from 80-bin histograms, (c) the Wasserstein-1 distance between sorted samples, and (d) moment gaps in mean, standard deviation, skewness, and excess kurtosis. Each metric emphasizes a different aspect of divergence: KS captures the worst-case CDF gap, JS captures average density mismatch, Wasserstein integrates the cost of transporting mass across the entire support (and is therefore sensitive to tails), and moment gaps localize the shift in distributional summaries.

Cross-market shift (LOBSTER-AAPL vs. FI-2010). Across all five features and all three dataset pairs, the resulting 15 two-sample tests are statistically significant at $p \ll 0.001$, confirming that distributions are non-identical even after per-dataset standardization. The *magnitude* of the shift, however, is highly heterogeneous across features. Resting depth is by far the most divergent quantity between LOBSTER-AAPL and FI-2010 (bid KS = 0.287, ask KS = 0.241), reflecting genuine structural differences between the deeply liquid US NASDAQ large-cap book and the comparatively thin Finnish Nordic books of the FI-2010 stocks. The mean depth gap of +0.42 IQR units (bid) and +0.33 IQR units (ask) confirms that LOBSTER books are systematically deeper in standardized terms, and the corresponding Wasserstein-1 distances (0.429 and 0.349) are the two largest in the entire metric suite. Spread shape exhibits moderate divergence (KS = 0.169), while return shape (KS = 0.078) and order imbalance (KS = 0.098) are broadly similar across markets (though as can be seen in Figure 2, the return tails of FI-2010 are considerably longer). The intuition is that aggregate price dynamics and order-flow balance reflect comparatively market-universal phenomena—short-horizon noise, mean-reversion, and the basic statistical signature of a double-auction market—while depth and spread structure are direct functions of liquidity provision and tick-size regimes that differ markedly between the two exchanges. Figure 2 visualizes these patterns: standardized return KDEs and tail CCDFs for the two markets sit nearly on top of one another, while the depth profiles diverge visibly across all ten levels. The spread panel confirms moderate misalignment, and the feature-level KS-statistic bar (lower right) ranks all five features by divergence magnitude, with both depth features at the top.

Cross-Market Domain Shift: FI-2010 vs LOBSTER-AAPL

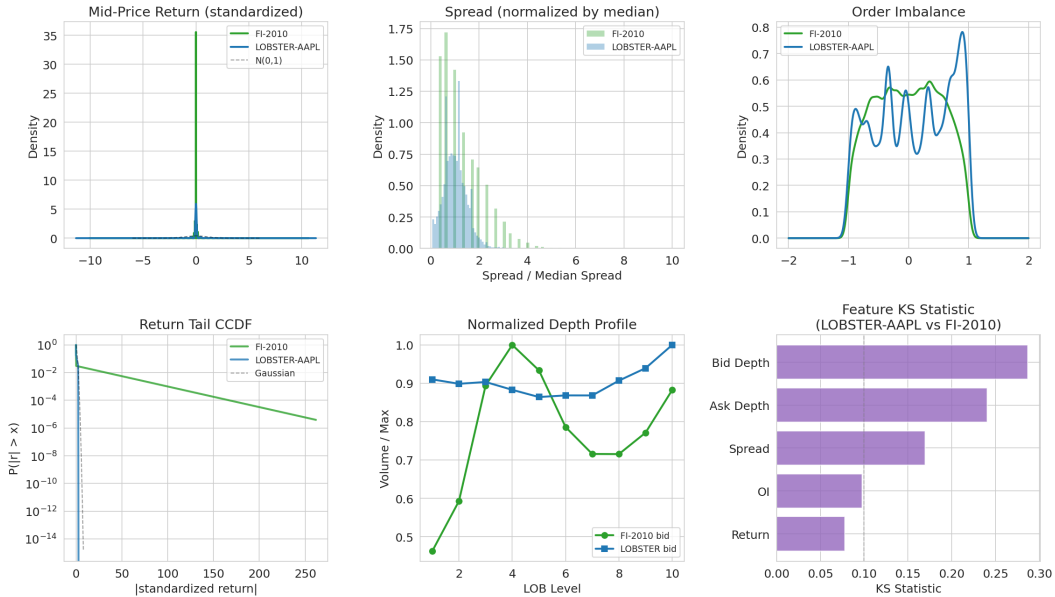


Figure 2: Cross-market distributional comparison between FI-2010 (training domain) and LOBSTER-AAPL (target domain). All features are standardized per-dataset before plotting. Returns and order imbalance overlap substantially across markets; spread shapes are moderately misaligned; depth profiles are the most divergent feature across all ten price levels.

Synthetic-vs-real shift (LOBS5). The LOBS5 generator was trained exclusively on US NASDAQ data, and a natural prior is therefore that its output should resemble LOBSTER more closely than FI-2010. The picture is more nuanced. Return shape and depth shape do behave as expected, with LOBS5 lying closer to LOBSTER ($KS_{ret} = 0.067$, bid/ask $KS_{depth} = 0.122/0.132$) than to FI-2010 ($KS_{ret} = 0.110$, bid/ask $KS_{depth} = 0.180/0.156$). Order imbalance and spread, however, invert this ordering: LOBS5 imbalance is closer in shape to FI-2010 ($KS = 0.042$) than to LOBSTER ($KS = 0.106$), and LOBS5 spread is closer to FI-2010 ($KS = 0.393$) than to LOBSTER ($KS = 0.484$). Spread is also the single most divergent feature between LOBS5 and either real dataset—the JS divergence reaches 0.456 against LOBSTER and 0.466 against FI-2010, both within a factor of 1.5 of the theoretical JS upper bound of $\ln 2 \approx 0.693$ —suggesting that the generator does not fully reproduce the discrete tick-clustering and tight-spread regime characteristic of US large-cap equities. Note also that LOBS5 depth has a uniformly *negative* mean gap against LOBSTER (-0.24 bid, -0.28 ask) but a *positive* gap against FI-2010 ($+0.17$ bid, $+0.06$ ask): LOBS5 books sit between the two real markets in terms of depth scale, shallower than LOBSTER but still deeper than FI-2010. The implication for our experiments is that LOBS5 cannot be relied on as a uniformly “closer-to-LOBSTER” source domain: it is closer for the features that drive directional return prediction (returns, depth) but further for the features that govern microstructure timing (spread, imbalance). Figure 3 illustrates these alignment patterns using QQ plots — LOBS5 return vs. LOBSTER-AAPL and LOBS5 order imbalance vs. FI-2010 — alongside the spread histogram (the largest residual gap) and depth profile (where LOBS5 tracks LOBSTER most closely). The Jensen–Shannon divergence heatmap in the lower-center panel summarizes the full feature-by-pair structure quantitatively.

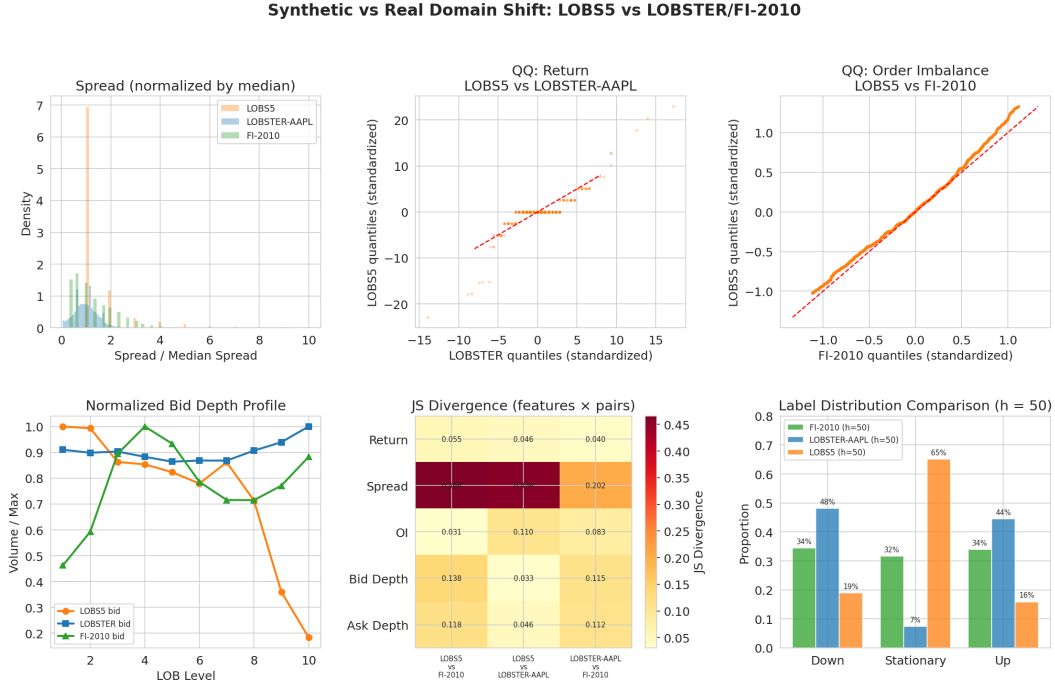


Figure 3: Synthetic-vs-real distributional comparison between LOBS5 and the two real datasets. LOBS5 returns and depth align more closely with LOBSTER (the market it was trained on); LOBS5 spread and order imbalance are unexpectedly closer to FI-2010. Spread is the largest residual gap between LOBS5 and either real dataset.

Divergence metrics comparison. Computing the full battery of metrics rather than relying on KS alone surfaces several cases of distinct *types* of shift that are operationally relevant for transfer learning. For LOBSTER-AAPL vs. FI-2010 returns, $KS = 0.078$ and $JS = 0.040$ both indicate small divergence, yet the Wasserstein-1 distance is 0.253 and the moment gaps are extreme: a skewness gap of -46.0 and an excess-kurtosis gap of $-7,106$ between the two return distributions. These extreme tail moments persist after per-dataset median/IQR standardization (which is moment-equivariant), and indicate that one or both return distributions contains rare extreme observations that move the higher moments by orders of magnitude even when the bulk of the density overlaps closely. KS and JS, which both weight the bulk of the distribution, miss this almost entirely; Wasserstein-1, which must transport probability across the extended tail at a real cost, partially captures it. Operationally, a model whose calibration is fit on the FI-2010 return distribution may therefore be systematically miscalibrated on LOBSTER tail events even though their bulk return distributions are nearly indistinguishable, with potential consequences for risk-aware downstream uses of the predictions. A second instructive disagreement appears between LOBS5 and LOBSTER on bid depth, where $JS = 0.033$ implies near-identical density shapes, yet

KS = 0.122 and Wasserstein-1 = 0.246 both flag a larger shift, and the mean moment gap of -0.245 IQR units confirms that LOBS5 depth is uniformly smaller than LOBSTER’s after median-centering. Density shapes are similar but the distributions are translated relative to one another. JS is insensitive to such translations whenever the supports remain largely overlapping, whereas KS and Wasserstein correctly identify the offset as a real divergence.

Table 1: Distributional divergence between dataset pairs across the five microstructure features used throughout this work, computed on per-dataset-standardized features ($n = 80,000$, seed 42). *KS* is the Kolmogorov–Smirnov statistic; *JS* is the Jensen–Shannon divergence (80-bin histograms); W_1 is the Wasserstein-1 distance on sorted samples; *Mean gap* is $\text{mean}(A) - \text{mean}(B)$ in IQR units. All KS tests are significant at $p \ll 0.001$. Bold values indicate the largest entry within each pair group; spread divergence dominates both LOBS5 comparisons, while depth dominates the LOBSTER-vs-FI-2010 cross-market gap.

Pair (A vs. B)	Feature	KS	JS	W_1	Mean gap
LOBSTER-AAPL vs. FI-2010	mid_return	0.078	0.040	0.253	+0.002
	spread	0.169	0.202	0.257	−0.240
	order_imbalance	0.098	0.083	0.088	+0.077
	total_bid_depth	0.287	0.115	0.429	+0.418
	total_ask_depth	0.241	0.112	0.349	+0.334
LOBS5 vs. LOBSTER-AAPL	mid_return	0.067	0.046	0.167	+0.004
	spread	0.484	0.456	0.485	+0.340
	order_imbalance	0.106	0.110	0.084	−0.047
	total_bid_depth	0.122	0.033	0.246	−0.245
	total_ask_depth	0.132	0.046	0.276	−0.275
LOBS5 vs. FI-2010	mid_return	0.110	0.055	0.146	−0.000
	spread	0.393	0.466	0.428	+0.099
	order_imbalance	0.042	0.031	0.043	+0.038
	total_bid_depth	0.181	0.138	0.233	+0.174
	total_ask_depth	0.155	0.118	0.150	+0.060

Implications for forecasting transfer. The above analysis predicts a specific pattern in the cross-market forecasting results of Section 4. Models that base their predictions primarily on aggregate return statistics or first-level imbalance signals should transfer with comparatively little degradation, since these features are the most market-universal in our measurements. Models that exploit fine-grained depth structure—as the convolutional stack of DeepLOB is explicitly designed to do (Section 1.2)—are most exposed to distribution shift, since depth profiles differ substantially between markets. The same reasoning further suggests that LOBS5-based adaptation is most likely to help in features that the generator already reproduces well (returns, depth) and least likely to help on the spread channel, where LOBS5 itself remains far from both real datasets. This corroborates the subpar results on finetuning DeepLOB using LOBS5 synthetic data (see Section 4.3).

3 Methodology

We organize our experiments into three tracks: (a) cross-market zero-shot and few-shot transfer, testing how a DeepLOB model trained on FI-2010 generalizes to LOBSTER with no adaptation and with small amounts of labeled target data; (b) synthetic-data adaptation, probing whether LOBS5 synthetic data can substitute for real LOBSTER labels via full supervised fine-tuning (SFT) and parameter-efficient LoRA; and (c) foundation-model evaluation, benchmarking Chronos-2 under zero-shot and fine-tuned conditions as a complement to the LOB-specific DeepLOB architecture.

3.1 Data Preprocessing and Label Generation

LOBSTER. Raw LOBSTER message files are replayed event-by-event to reconstruct L2 order book snapshots at the top 10 price levels on each side. The data is split 70/30 by time for few-shot fine-tuning experiments. Labels at each snapshot are generated using Equation (5) with an absolute mid-price change threshold $\alpha = 10^{-5}$ USD, chosen to produce an approximately balanced three-class label distribution over the single trading session.

LOBS5. The same L2 reconstruction pipeline is applied to the LOBS5 synthetic message streams. Because each message file corresponds to an independent simulation run, z-score normalization statistics are computed within each segment to prevent cross-segment contamination; segment-aware windowing ensures that no input window straddles two independent simulations.

FI-2010. The first 40 columns of the FI-2010 feature matrix (the raw price–volume snapshot) are extracted and aligned to the LOBSTER-compatible schema. Pre-computed labels at horizons $k \in \{10, 20, 50\}$ are used directly. The standard 7-day training / 3-day test split is used for in-domain experiments; for cross-market pretraining, the full 10-day dataset is used.

All three datasets share the same input format: a window of $T = 100$ consecutive snapshots, each a 40-dimensional vector, yielding a 100×40 input tensor (see Equation (3)).

3.2 DeepLOB Training and Cross-Market Evaluation

We train DeepLOB (Section 1.2) from scratch on FI-2010 for each horizon $k \in \{10, 20, 50\}$. Training uses the Adam optimizer with learning rate 10^{-2} ($\epsilon = 1.0$), batch size 32, and a patience-20 early-stopping criterion on balanced accuracy, for a maximum of 100 epochs. The best-balanced-accuracy checkpoint is saved and used as the starting point for all subsequent adaptation experiments.

Zero-shot cross-market performance is obtained by applying the FI-2010 checkpoint directly to LOBSTER-AAPL without any weight updates.

3.3 Few-Shot Fine-Tuning on the Target Market

Starting from the FI-2010 checkpoints, we fine-tune on subsets of LOBSTER-AAPL training data at fractions $\{1\%, 5\%, 10\%, 20\%, 50\%\}$ of the 70% training split. Two fine-tuning modes are tested:

- **Full:** all model weights are updated.
- **Head-only:** the convolutional stack and Inception module are frozen; only the LSTM and classifier are updated.

Both modes use Adam with learning rate 10^{-4} , maximum 30 epochs, and patience 7. Each (fraction, mode) configuration is run with 3 random seeds to report mean $\pm$ standard deviation. Complete results are presented for horizon $k = 10$.

3.4 Synthetic-Data Adaptation with LOBS5

Full SFT and LoRA fine-tuning. We fine-tune the FI-2010-pretrained DeepLOB checkpoints on LOBS5 synthetic data (2.51M reconstructed events across 5,056 independent simulation segments from GOOG and INTC generator runs). Two adaptation regimes are compared.

Full SFT updates all parameters with Adam (learning rate 10^{-4} , weight decay 10^{-5} , batch size 64, gradient clipping at norm 1.0) for a maximum of 15 epochs with patience 4.

LoRA inserts rank-8 adapter matrices (scaling factor $\alpha/r = 16/8 = 2$) into the LSTM input and hidden weight matrices (`weight_ih_10`, `weight_hh_10`) and the classifier projection. The convolutional stack and Inception module remain frozen, yielding 5,912 trainable parameters (8.8% of total), trained with Adam at learning rate 5×10^{-4} .

After adaptation on LOBS5, both models are evaluated on the LOBSTER-AAPL 30% test split used for zero-shot evaluation.

Targeted segment selection. For Chronos-2 adaptation we additionally implement a targeted data-selection step. For each of the 360 LOBS5 synthetic segments, 4,000 events are sampled and four feature distributions (mid-price returns, order imbalance, spread, depth) are compared against LOBSTER-AAPL using Wasserstein-1 distance and the KS statistic. A composite similarity score is computed as a weighted sum of normalized per-feature distances:

$$s = 0.25 \hat{W}_{\text{ret}} + 0.20 \hat{W}_{\text{oi}} + 0.15 \hat{W}_{\text{spr}} + 0.10 \hat{D}_{\text{ret}}^{\text{KS}} + 0.10 \hat{D}_{\text{oi}}^{\text{KS}} + 0.10 \hat{W}_{\text{dep}} + 0.10 \hat{\Delta}_{\mu}, \quad (6)$$

where hats denote min-max normalization across all segments and $\hat{\Delta}_{\mu}$ is the normalized mean-gap between the segment and LOBSTER-AAPL. Lower s indicates higher similarity to the target. The top-30 most similar segments are selected and augmented $12 \times$ (price jitter, volume scaling, additive noise) to yield 169,560 synthetic windows, which are merged with the FI-2010 training set to form the augmented corpus used for Chronos-2 fine-tuning.

3.5 Foundation-Model Baseline: Chronos-2

Chronos-2 [1] is an encoder-only time-series foundation model with approximately 120M parameters pretrained on a large and diverse collection of real-world time series. We adapt it for LOB mid-price movement classification

by treating the task as a point-forecasting problem: the model receives $T = 100$ LOB snapshots as context and produces a predictive distribution over the next k steps; the median quantile ($q = 0.5$) forecast is compared against the current mid-price using the same $\alpha = 10^{-5}$ threshold to produce a three-class label.

Three conditions are evaluated on LOBSTER-AAPL:

1. **Zero-shot:** pretrained weights, no LOB-specific training.
2. **FI-2010 fine-tuned:** 994 sliding-window segments from FI-2010 (context length 512, learning rate 10^{-4} , cosine schedule with 10% warmup, 3 epochs).
3. **FI-2010 + filtered LOBS5:** FI-2010 merged with the top-30 AAPL-similar LOBS5 segments (augmented $12\times$, 1,079 total segments), same fine-tuning protocol.

Transfer is evaluated on 1,000 held-out LOBSTER-AAPL test windows.

3.6 Evaluation Metrics

Our primary metric is **balanced accuracy** (arithmetic mean of per-class recall), which treats all three movement classes equally and is robust to the class imbalance typical in LOB data. We also report **weighted F1**, **macro F1**, and **Cohen’s κ** ($\kappa \approx 0$ indicates chance level; negative values denote below-chance performance). All metrics are computed on the held-out LOBSTER-AAPL test split unless stated otherwise.

4 Results

4.1 DeepLOB: Collapse Under Zero-Shot Cross-Market Transfer

DeepLOB achieves strong in-domain performance on FI-2010, with balanced accuracies of 0.68, 0.61, and 0.72 for $k = 10, 20$, and 50 respectively. Applied zero-shot to LOBSTER-AAPL, performance collapses to near-random: balanced accuracy falls to 0.302, 0.306, and 0.334 for the three horizons — a drop of roughly 30–38 percentage points. Weighted F1 falls to 0.228, 0.141, and 0.104, and Cohen’s κ is at or below zero across all horizons ($\kappa \approx -0.03$ to 0.00), confirming that the model is performing at or worse than chance on the target market.

This collapse is precisely what the domain-shift analysis of Section 2 predicts. The depth features that diverge most sharply between markets (bid/ask KS = 0.29/0.24) are the same features that DeepLOB’s convolutional stack is explicitly designed to exploit. When depth structure changes, the learned spatial filters cease to be informative and the model degenerates toward predicting the dominant class. Figure 4 shows the resulting confusion matrices: predictions are concentrated in a single column, with the two minority classes almost entirely missed.

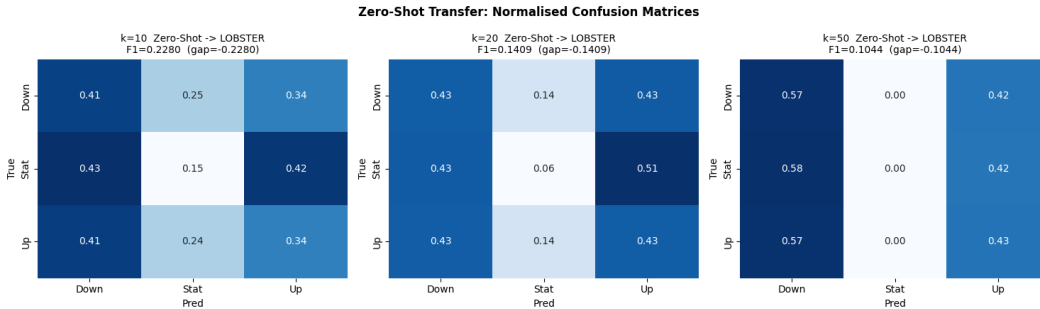


Figure 4: Normalized confusion matrices for zero-shot DeepLOB transfer from FI-2010 to LOBSTER-AAPL at horizons $k \in \{10, 20, 50\}$. Predictions collapse to the majority class, confirming that the model’s depth-based representations do not transfer across markets.

4.2 Few-Shot Fine-Tuning: Rapid Recovery with 1% Real Data

Despite the severity of the zero-shot collapse, fine-tuning on a small fraction of labeled LOBSTER data restores performance remarkably quickly. At horizon $k = 10$, using just 1% of the LOBSTER training split raises balanced accuracy from 0.302 to 0.526 — a gain of 22 percentage points that more than halves the gap between zero-shot and the best performance achievable with full data. Performance continues to improve with more data but with strong diminishing returns (see Table 2).

Full fine-tuning of all weights consistently outperforms the head-only variant, which achieves only 0.463 balanced accuracy at 1% data. This indicates that the convolutional depth-sensing layers — the very components that fail under distribution shift — must be re-calibrated along with the recurrent layers to successfully adapt to the new market. Table 2 summarizes the full $k = 10$ sweep.

Table 2: DeepLOB balanced accuracy on LOBSTER-AAPL at horizon $k = 10$ as a function of the fraction of LOBSTER training data used for fine-tuning, for full fine-tuning (all weights) and head-only fine-tuning (Conv + Inception frozen). Mean over 3 seeds. Zero-shot (0%) is the FI-2010 checkpoint applied directly without adaptation.

Data fraction	Full fine-tune	Head-only
0% (zero-shot)	0.302	0.302
1%	0.526	0.463
5%	0.546	0.485
10%	0.554	0.490
20%	0.562	0.498
50%	0.678	0.552

The diminishing-returns pattern mirrors what we observe in the in-domain setting: when varying FI-2010 training set size, both DeepLOB and a LOBTransformer baseline reach approximately 90% of full-data performance with only 5–20% of training data (depending on horizon). Diminishing returns under LOB fine-tuning thus appear to be a structural property of these tasks, not a quirk of the cross-market setting.

4.3 Synthetic-Data Adaptation: LOBS5 Fails to Bridge the Gap

Given the effectiveness of real target labels, a natural question is whether high-quality synthetic data can substitute for them. Table 3 reports the results of fine-tuning the FI-2010 DeepLOB checkpoint on LOBS5 synthetic data via full SFT and LoRA, evaluated on LOBSTER.

Table 3: DeepLOB balanced accuracy on LOBSTER-AAPL after synthetic fine-tuning on LOBS5, compared to the zero-shot baseline. Δ columns give the gain over zero-shot. Cohen’s $\kappa \approx 0$ throughout all conditions, indicating near-chance performance.

k	Zero-shot	SFT Δ	LoRA Δ
10	0.311	+0.030	+0.021
20	0.305	+0.031	+0.022
50	0.331	+0.001	+0.031

The gains are marginal at every horizon and within estimation noise. Notably, SFT achieves balanced accuracy of 0.71–0.73 on the held-out LOBS5 validation set, confirming that the models do adapt to the synthetic distribution — the representation simply does not transfer to real market data.

This failure is consistent with the distributional analysis of Section 2.1. LOBS5 spread divergence from LOBSTER is extreme ($KS = 0.484$, $JS = 0.456$): spread is precisely the feature the synthetic generator fails to replicate despite being trained on US NASDAQ data. A further compounding factor is the temporal regime gap: LOBS5 was generated to resemble 2022 market conditions, while LOBSTER captures 2012 AAPL trading. The dramatic response of DeepLOB to even 1% of real 2012 LOBSTER data (Section 3.3) establishes that regime-correct data is the bottleneck. The near-equivalence of full SFT and LoRA outcomes further suggests the failure is one of distributional mismatch rather than overfitting: adding more trainable parameters provides no benefit when the source and target distributions diverge on the critical features. Figure 5 visualizes the near-identical LOBSTER performance of all three conditions.

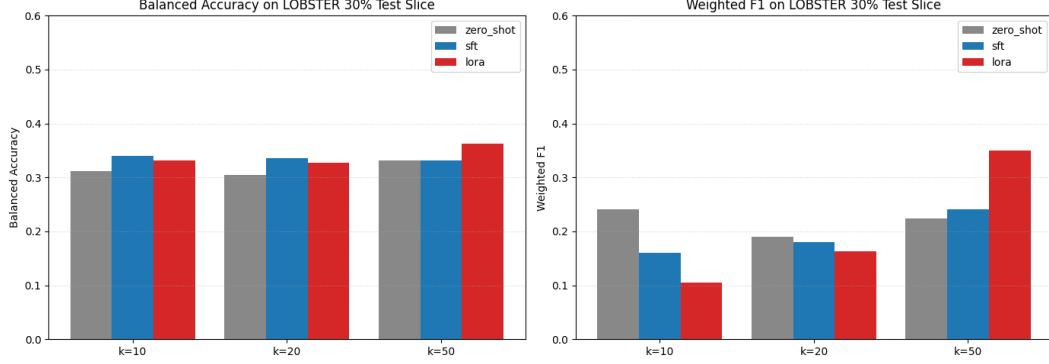


Figure 5: *Balanced accuracy and weighted F1 on LOBSTER-AAPL for zero-shot, LOBS5 SFT, and LOBS5 LoRA fine-tuned DeepLOB, across horizons $k \in \{10, 20, 50\}$. All conditions remain near chance (≈ 0.33 balanced accuracy). High LOBS5 validation performance confirms that models fit the synthetic distribution but that the fit does not transfer to real market data.*

4.4 Foundation-Model Transfer: Chronos-2

Chronos-2 offers a complementary perspective on the cross-market transfer problem. As a foundation model pretrained on diverse time-series corpora, it does not exploit LOB depth structure explicitly and is therefore less exposed to the specific source of distribution shift that damages DeepLOB.

Zero-shot performance. Under zero-shot conditions, Chronos-2 achieves macro-F1 of 0.322, 0.313, and 0.301 on LOBSTER-AAPL for $k = 10, 20$, and 50 respectively (accuracy ≈ 0.44 across horizons). These numbers are meaningfully higher than DeepLOB’s zero-shot macro-F1 (≈ 0.24), suggesting that broad pretraining on diverse temporal data does capture some market-universal patterns that transfer without any LOB-specific training.

Fine-tuning effects. Adapting Chronos-2 to FI-2010 produces slight *negative* transfer: macro-F1 drops to 0.299, 0.296, and 0.303 for $k = 10, 20, 50$, consistent with FI-2010 being too small ($\approx 255K$ events) and too domain-specific to beneficially update a 120M-parameter model. Adding the merged FI-2010 + LOBS5 corpus recovers marginally to 0.302, 0.298, and 0.302 — statistically indistinguishable from zero-shot. Even the targeted pipeline (top-30 AAPL-similar LOBS5 segments, $12\times$ augmented) achieves a best macro-F1 of only 0.327 with accuracy 0.455, an improvement of just $\approx +0.015$ over the zero-shot baseline. Table 4 summarizes all conditions.

Table 4: *Chronos-2 macro-F1 on LOBSTER-AAPL across training conditions and prediction horizons. “Targeted LOBS5” uses the top-30 most AAPL-similar segments scored by the composite metric of Equation (6), augmented $12\times$, combined with LOBSTER-AAPL training windows.*

Condition	$k = 10$	$k = 20$	$k = 50$	Mean
Zero-shot	0.322	0.313	0.301	0.312
FI-2010 fine-tuned	0.299	0.296	0.303	0.299
FI-2010 + LOBS5	0.302	0.298	0.302	0.301
Targeted LOBS5 ($k = 10$)	0.327	—	—	—

The Stationary class is predicted with $F1 = 0.00$ under all conditions, reflecting class imbalance and the model’s tendency to forecast directional movement. Directional classes show moderate performance in the best condition ($F1 = 0.46$ Down, 0.53 Up). Figure 6 shows the confusion matrix for the targeted fine-tuning condition.

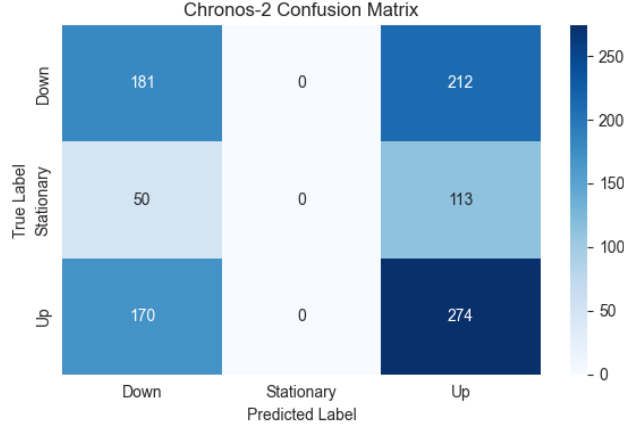


Figure 6: Normalized confusion matrix for Chronos-2 fine-tuned with targeted AAPL-similar LOBS5 segments on LOBSTER-AAPL. The Stationary class is entirely missed; directional classes (Down, Up) show F1 of 0.46 and 0.53 respectively.

4.5 Discussion

Taken together, the results reveal several interconnected patterns about domain shift in LOB forecasting.

Architecture determines transfer vulnerability. DeepLOB’s convolutional stack is designed to extract depth-level spatial features — precisely the features that diverge most sharply between markets (depth KS = 0.24–0.29). As a result, it collapses under zero-shot transfer while Chronos-2, which imposes no LOB-specific inductive bias, achieves a higher zero-shot floor. The flip side is that DeepLOB responds dramatically to targeted re-calibration: full fine-tuning allows the depth-sensing layers to adapt to the new market structure, producing a 22-point balanced-accuracy gain from just 1% of real LOBSTER data.

Effective fine-tuning requires regime-correct data. The contrast between the 1% real-data result and the LOBS5 synthetic result is stark. A small set of real LOBSTER windows from 2012 is worth far more than millions of synthetic windows generated from a 2022 GOOG distribution. This is not a failure of synthetic data generation in principle; it is a consequence of the temporal regime gap and the specific distributional mismatch in spread (KS = 0.484) — the feature LOBS5 is least able to replicate. A synthetic generator that reproduces the target market regime and spread dynamics would likely close this gap substantially, given how sensitive DeepLOB is to even a tiny amount of regime-correct supervision.

Foundation models provide a better zero-shot baseline but resist fine-tuning. Chronos-2’s higher zero-shot macro-F1 (≈ 0.31 – 0.32 vs. ≈ 0.24 for DeepLOB) suggests that broad pretraining does capture some market-universal temporal features. However, fine-tuning Chronos-2 on the small, domain-specific FI-2010 dataset degrades performance slightly, and targeted LOBS5 augmentation yields only marginal gains. The same broad pretraining that provides a robust zero-shot baseline makes domain-specific adaptation harder: small, noisy LOB datasets are insufficient to meaningfully update a 120M-parameter model. Moreover, the results indicate the exposing this foundation model to structured LOB data does not meaningfully improve performance if that data remains out of distribution relative to cross-market shift, leaving us in the same position as with DeepLOB in the setting without access to target data. Better adaptation strategies remain an open direction.

Deployment implications. The few-shot results demonstrate that even a small amount of in-distribution data is highly valuable for LOB-specific models. In live trading, however, a model must be ready before market open: there is no opportunity to collect labeled windows from the current session and fine-tune before the first trade. This creates a fundamental tension. Practically, it motivates either (a) pre-adapting the model on historical data from the target market before each session, (b) developing synthetic generators that are regime-correct and reproduce the target spread distribution, or (c) favoring foundation-model architectures whose broader pretraining provides a more robust zero-shot starting point while being adapted offline. The marginal gains from LOBS5 adaptation ($\approx +0.015$ macro-F1 for Chronos-2, $\approx +0.03$ balanced accuracy for DeepLOB SFT) indicate that current synthetic LOB data cannot be applied out of the box to strengthen performance under cross market shift (generating data explicitly conditioned on the target distribution remains an open research direction).

5 Next steps

Our results suggest several concrete directions for future work, three of which target the data side of the cross-market transfer problem and one of which targets the model.

Target-conditioned synthetic generation. The analysis of Section 2.1 and the results of Section 4.3 point to a sharp limitation of LOBS5 as a transfer source: it fits its training distribution well but diverges from any specific target market on features (notably spread) that turn out to be essential for forecasting. A natural next step is to move from *selecting* synthetic samples that happen to resemble the target—as our composite-score pipeline in Section 3.4 does—to actively *generating* them, by conditioning or fine-tuning a generative model such as LOBS5 on a small calibration sample from the target market. Concretely, this might take the form of brief continued pre-training of LOBS5 on a single day of target LOBSTER data, target-conditioned message sampling, or distributional matching via guidance during generation. Given how sharply DeepLOB responds to even 1% of real target data, a generator that closes the spread and depth gap by a moderate margin could materially improve cross-market transfer at scales of synthetic data that real labels cannot reach.

Pre-session adaptation on adjacent historical data. Our few-shot experiments establish that small quantities of regime-correct target data yield large gains, but the live-deployment constraint of being ready before market open rules out fine-tuning on the current session. A practical compromise is to fine-tune on the most recent available trading day(s) from the target market and then deploy. Open questions include how performance degrades as the gap between training and test sessions grows, whether day-of-week effects or pre-event regime breaks weaken the benefit, and whether combining several adjacent days outperforms reliance on only the most recent one. We were unable to investigate this in the present work because publicly available LOBSTER data is limited to a single trading day; real market participants would likely have direct access to LOB data.

Synthetic data for in-market robustness. A distinct use of synthetic data that we did not examine is the strengthening of in-distribution behavior rather than cross-distribution transfer, particularly robustness to tail events such as flash crashes, news shocks, and halts that are rare even in multi-year datasets. The DSLOB benchmark [3] explicitly constructs synthetic LOB data with and without labeled market-stress scenarios for precisely this purpose, providing a testbed in which controlled distributional shifts can be evaluated against in-sample performance. An interesting extension of our work would be to augment FI-2010 or LOBSTER training with synthetic stress segments—drawn either from DSLOB or from a generator conditioned on shock-like statistics—and measure robustness to tail events while monitoring nominal-regime accuracy. The success criterion here differs from the rest of this paper: the goal is not to close a distribution gap but to expand the support of the training distribution into rare but operationally important regions.

Representation-level invariance. A final direction targets the model rather than the data. Section 2.1 identifies the dominant axis of cross-market shift as a translation in the standardized depth distribution, and Section 3.3 shows that the depth-sensing convolutional layers are exactly what must be re-calibrated to recover performance. This suggests that learning representations invariant to such nuisance dimensions—via domain-adversarial training across markets, instance or feature normalization tailored to LOB scale, or contrastive objectives that align cross-market embeddings—could reduce zero-shot fragility without requiring any target labels. Unlike the data-side directions above, this approach attacks the underlying source of the failure mode rather than working around it, and we view it as a complement to rather than a substitute for the synthetic-data and few-shot strategies.

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